Marcelo HURTADO

BSc Bioengineering – Biotechnological Engineering
PhD student in Computational Biology & Bioinformatics

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EDUCATION

Oct 2023

PhD student ★3 ★4 ★5

- Paul Sabatier University, Toulouse, France
- Cancer Research Center of Toulouse (CRCT)
- ***Team:*** NetB(IO)^2 | ***Supervisor:*** Vera Pancaldi
- ***Project:*** Reinforcement learning approaches to control the tumor microenvironment

Current achievements:

- Develop a multiscale ABM of NLC-CLL differentiation using the **Physicell** software (Github repository).
- Develop a ***Python*** and ***Bash*** pipeline for Physicell model exploration (including parameter exploration, sensitivity analysis and genetic algorithms) for running in ***HPC arquitectures*** (Github repository).
- Apply basic reinforcement learning algorithms into the NLC-CLL model to evaluate cancer cell concentration
- Develop an advanced machine learning framework in ***R*** to extract clinically relevant features from bulk RNAseq data (***pipeML***)

Jan 2017 – Dec 2023

B.Sc Bioengineering - Mention in Biotechnology

- Universidad de Ingenieria y Tecnologia UTEC Lima, Peru.
- ***Courses***: Synthetic Biology, Biomedical signals, Medical robotics, bioprocesses, biofluids mechanics.

Aug 2021 – Dec 2021

Exchange Program | Biotechnology and Biomedical Engineering

- Instituto de Estudios Superiores Tecnologicos de Monterrey - Monterrey, Mexico.
- ***Courses:*** Genetic engineering, neuroengineering, biomedical imaging

RESEARCH EXPERIENCE

Jan 2021 – Aug 2021

Visiting PhD Student | Eindhoven University of Technology (TU/e). ★9

- ***Department***: Biomedical Engineering | ***Supervisor:*** Federica Eduati
- ***Tasks***: Personalisation of mechanistic Boolean models through integration of transcriptomics data.
- ***Achievements***: Improvement and development of a R package for cell-cell communication from bulk RNA-seq based on kernel and Monte-Carlo algorithms (**RaCiNG**) .

Apr 2023 – Oct 2023

Bioinformatics Engineer | Cancer Research Center of Toulouse (CRCT)

- ***Team:*** NetB(IO) | ***Principal Investigator:*** Vera Pancaldi
- ***Tasks:*** Analisis of transcriptomics data to evaluate patient profiles of early and late stage NSCLC patients.
- ***Achievements:*** Develop a novel framework for characterizing TME patient profiles by constructing transcriptional regulatory networks (TRNs) based on inferred transcription factor (TF) activity and cell type deconvolution from bulk RNA-seq data (***CellTFusion***).

<u>**Hurtado M**</u> et al. (2024). Transcriptomics profiling of the non-small cell lung cancer microenvironment across disease stages reveals dual immune cell-type behaviors. Front. Immunol. 15:1394965. doi: 10.3389/fimmu.2024.1394965

Jan 2022 – March 2022

Aug 2022 – Feb 2023

Intern of Bioinformatics | Cancer Research Center of Toulouse (CRCT) ★6 ★7

- ***Team***: NetB(IO)^2 | ***Principal Investigator:*** Vera Pancaldi
- ***Tasks***: Analysis of tumor cell composition on NSCLC patients using computational tools based on transcriptomics data.
- ***Achievements***: Develop of an algorithm that integrates a combination of multiple first or second-generation deconvolution methods and several cell type signatures to ensure robust and accurate profiling of cell composition starting from bulk RNAseq data (***multideconv***).

★ *Awards and achievements*

Jan 2021
Aug 2021

Research Assistant | Universidad de Ingenieria y Tecnologia UTEC Lima, Peru.

- ***Department***: Bioengineer Department | ***Supervisor:*** Dr. Alberto Donayre
- ***Tasks***: Design of a bioactive polymer based on self-assembling and antimicrobial peptides for wound applications.
- ***Achievements*:** Propose a genetic cassette for the production of self-assembling peptides with antimicrobial properties via the implementation of genetic engineering techniques and cloning methods (***Gibson assembly***, ***restriction enzymes***, ***golden gate***).

PUBLICATIONS

Hurtado, M., & Pancaldi, V. (2026). CellTFusion: A transcriptional regulatory network framework for the identification of functional multicellular states from bulk RNA-seq data. bioRxiv. https://doi.org/10.64898/2026.06.30.735682

Gobbini E., Duplouye P., **Hurtado M.**, ... Pancaldi V., Valladeau-Guilemond J. (2026). Specific dendritic cells spatial organization is associated to ICB Response in Non Small Cell Lung Cancer (NSCLC). bioRxiv. https://doi.org/10.64898/2026.05.04.720587 (Gobbini E., Duplouye P. and Hurtado M. contributed equally to this work)

Hurtado M., Pancaldi V. (2026). A new pipeline for cross-validation fold-aware machine learning prediction of clinical outcomes addresses hidden data-leakage in omics based predictors. bioRxiv 2026.03.12.711429. https://doi.org/10.64898/2026.03.12.711429

Marku, M.*, Chenel, H.*, ..., **Hurtado M.**, et al. Data driven network inference and longitudinal transcriptomics unveil dynamic regulation in Chronic Lymphocytic Leukaemia models. npj Syst Biol Appl (2026). https://doi.org/10.1038/s41540-025-00645-4

Bertin A*, Bucher E*, ..., **Hurtado M**, et al. (2025). PhysiGym: bridging the gap between the Gymnasium reinforcement learning application interface and the PhysiCell agent-based model software. bioRxiv 2025.09.18.677030; doi: https://doi.org/10.1101/2025.09.18.677030

Hurtado M., et al. (2025). multideconv – Integrative pipeline for cell type deconvolution from bulk RNAseq using first and second generation methods. bioRxiv. https://doi.org/10.1101/2025.04.29.651220

Hurtado M*, Khajavi L*., et al. (2024). Transcriptomics profiling of the non-small cell lung cancer microenvironment across disease stages reveals dual immune cell-type behaviors. Front. Immunol. 15:1394965. doi: 10.3389/fimmu.2024.1394965

ORAL PRESENTATIONS

12/05/2026
14/05/2026

Estimating tumour microenvironment cellular states from bulk RNA-seq reveals biomarkers of clinical outcome in Non Small Cell Lung Cancer (NSCLC) across disease stages
EACR – Cancer Genomics, Multiomics and Computational Biology - Essen, Germany.

01/12/2025
04/12/2025

CellTFusion: A novel approach to unravel cell states via cell type deconvolution and TF activity estimated from bulk RNAseq data identifies clinically relevant cell niches
Bioinformatics and Computational Biology Conference - Napoli, Italy.

26/11/2025
28/11/2025

Estimating tumour microenvironment cellular states from bulk RNAseq produces biomarkers of clinical outcome across stages
21èmes Journées annuelles du Cancéropôle GSO - Agen, France.

20/07/2025
24/07/2025

CellTFusion: A novel approach to unravel cell states via cell type deconvolution and TF activity estimated from bulk RNAseq data identifies clinically relevant cell niches
European Conference on Computational Biology - ISMB/ECCB 2025 - Liverpool, UK.

07/11/2024

"CellTFusion: Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression"
Journee Bioinfo/Biostat GenoToul - Toulouse, France.

26/11/2023
01/12/2023

"Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression"
EMBO Workshop | Computational models of life: From molecular biology to digital twins
Sant Feliu de Guíxols, Spain.

★ *Awards and achievements*

22/11/2023
24/11/2023
"Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression"
19th Annual Meeting of the Canceropole GSO - Arcachon, France. ★1

25/09/2023
29/09/2023
"Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression"
Institut Curie | Computational Systems Biology of Cancer 6th edition - Paris, France. ★2

POSTER PRESENTATIONS

2025
Estimating NSCLC tumour microenvironment cellular states from bulk RNAseq produces biomarkers of clinical outcome across stages
World Conference on Lung Cancer (WCLC) 2025 - Barcelona, Spain.

2025
CellTFusion: A novel approach to unravel cell states via cell type deconvolution and TF activity estimated from bulk RNAseq data identifies cell niches potentially predictive of cancer progression
JOBIM 2025 - Bordeaux, France

2024
Transcriptional regulatory networks unravel cell states from immune cell type deconvolution and uncovers cell niches predictive of cancer progression
1st edition Young Scientist Cancer Congress - Toulouse, France

2023
Profiling of the Non-Small Cell Lung Cancer (NSCLC) Microenvironment Across Disease Stages
EACR Defence is the Best Attack: Immuno-Oncology Breakthroughs 2023 - Barcelona, Spain.

2023
Profiling of the Non-Small Cell Lung Cancer (NSCLC) Microenvironment Across Disease Stages
The Festival of Genomics & Biodata 2023 - London, UK.

2022
Multi-omics Profiling of the Non-Small Cell Lung Cancer (NSCLC) Microenvironment Across Disease Stages and Gender
Cancéropôle Grand Sud-Ouest 18th Annual Meeting - Montpellier, France.

REVIEWING AND EDITING ACTIVITIES

Review Commons - Reviewer

Dugourd, A., et al. (2024). Modeling causal signal propagation in multi-omic factor space with COSMOS. bioRxiv. https://doi.org/10.1101/2024.07.15.603538

Review Commons - Reviewer

Trimbour, R., Ramirez Flores, R. O., Saez-Rodriguez, J., & Cantini, L. (2026). Modelling multicellular coordination by bridging cell-cell communication and intracellular regulation through multilayer networks. bioRxiv. https://doi.org/10.64898/2026.01.20.700561

Clinical Cancer Research - Reviewer

Anders, P., et al (2026). ADC target profiling in NSCLC: Generalizable AI separates TROP-2 and cMET phenotypes. Clinical Cancer Research, 32(13), 2765-2786. https://doi.org/10.1158/1078-0432.CCR-25-4513

LEADERSHIP & OUTREACH

Workshop organization - Innovations in immuno-oncology: from data to therapeutic insights

Team leader - iGEM Desing League 2021 Synthetic Biology competition. ★8

Project *LECCHAIN:* Improving vaccine thermal tolerance and stability of SARS-CoV-2 antigen using plant lectins.

Press release:

- Centro Bio
- Press coverage
- Peru21
- ElComercio

★ *Awards and achievements*

COURSES AND WORKSHOPS

02/12/2024
06/12/2024
Health Data Challenge 2024: Multimodal data integration to quantify tumor heterogeneity in cancer research.
Aussois, France.

29/06/2024
05/07/2024
14th Summer school on medicines.
University of Sao Paulo, Ribeirao Preto Medical School.

21/04/2024
26/04/2024
Computational Systems Biology for Complex Human Disease: from static to dynamic representations of disease mechanisms.
Wellcome Genome Campus, Hinxton, United Kingdom.

26/11/2023
01/12/2023
EMBO Workshop. Computational models of life: From molecular biology to digital twins.
Sant Feliu de Guíxols, Spain.

25/09/2023
29/08/2023
6th course on Computational Systems Biology of Cancer: models of data, data for models.
Institut Curie, Paris, France.

SUPERVISION

2026 - Rajath PRABHU MANJESHWAR - **Masters M2 program (6 months)**
Project: *Investigating CLL Cell plasticity through Co-Culture systems and Agent Based Modelling.*
Cancer Research Center of Toulouse (CRCT)

2026 - Mariajose CAZORLA - **International program (3 months)**
Project: *Transcriptomics Profiling of the Tumor Microenvironment in KRAS Mutated Patients with Non-Small Cell Lung Cancer.*
Cancer Research Center of Toulouse (CRCT)

2024 - Francesco MASSAINI - **Masters M2 program (6 months)**
Project: *Profiling the Immune Landscape: Multi-Omics analysis of the Tumor Microenvironment in Non-Small Cell Lung Cancer.*
Cancer Research Center of Toulouse (CRCT)

AWARDS AND ACHIEVEMENTS

1. Prix de la Fondation Silab Jean Paufique
2. Best oral presentation award
3. Campus France - France Excellence Eiffel PhD scholarship
4. Toulouse Foundation Cancer Sante - PhD scholarship
5. CARe Graduate School Univ. Paul Sabatier - Doctoral School Fellowship
6. CARe Graduate School Univ. Paul Sabatier - M2 internship fellowship
7. Research for Peruvian Undergraduates (REPU) program - International stage fellowship
8. iGEM Design League - Gold medal, Best Human Practices, Best aligned with SGD
9. ERASMUS+ Fellowship

REFERENCES

Dr. Vera **PANCALDI**
Principal Investigator (NetBIO)
Cancer Research Center of Toulouse
vera.pancaldi@inserm.fr

Dr. Julien **MAZIERES**
Pulmonology Department, Larrey Hospital, University Hospital of Toulouse,
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Dr. Elisa **GOBBINI**
Centre de Recherche en Cancérologie de Lyon
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Awards and achievements